

TELCO CUSTOMER RETENTION & CHURN ANALYSIS REPORT

Data Science & Analytics Internship Task 2

Future Interns 2026

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1. Executive Summary

This report presents a comprehensive analysis of customer churn and retention behaviour for a telecommunications company. The analysis is based on 7,043 customer records which includes contract types, payment methods, internet service plans, and tenure periods. The goal is to identify the key drivers of customer churn and provide recommendations to help the business improve retention and reduce revenue loss.

1.1 Key Performance Indicators

Total Customers	Churned Customers	Churn Rate	Retention Rate	Churned Revenue
7,043	1,869	26.54%	73.46%	\$2,862,927

Out of 7,043 customers, 1,869 have churned representing a churn rate of 26.54%. This means 1 in 4 customers is leaving the platform. The business has lost \$2,862,926.90 in revenue from churned customers, out of a total revenue base of \$16,056,168.70. Reducing churn by even a few percentage points would have a significant impact on the profit.

2. Churn Analysis Dashboard

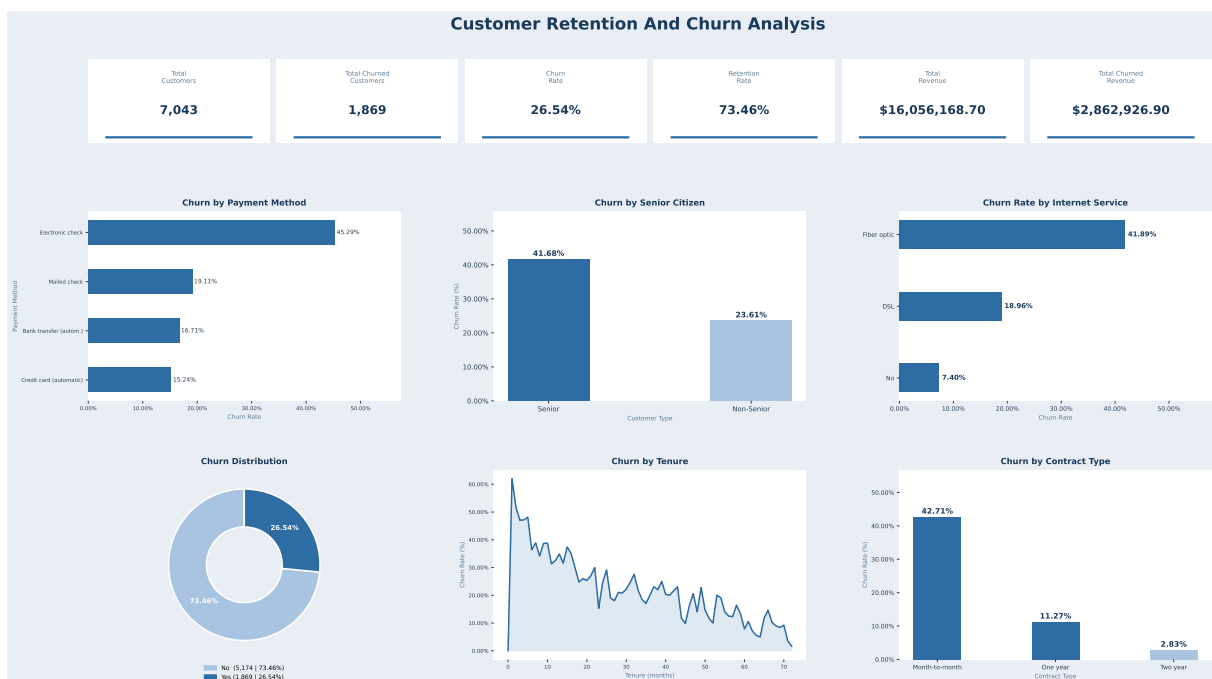


Figure 1: Telco Customer Retention & Churn Analysis Dashboard for 7,043 Customers

3. Churn by Contract Type

Contract type is the single strongest predictor of churn in this dataset. The analysis reveals a clear and dramatic pattern:

- **Month-to-month contracts:** 42.71% churn rate by far the highest
- **One year contracts:** 11.27% churn rate significantly lower
- **Two year contracts:** Only 2.83% churn rate the most loyal customers

Key Finding: Customers on month-to-month contracts are over 15 times more likely to churn than those on two-year contracts. Contract length is the most powerful retention lever available to the business.

4. Churn by Payment Method

Payment method reveals meaningful differences in customer loyalty and engagement:

- **Electronic check:** 45.29% churn rate highest of all payment methods
- **Mailed check:** 19.11% churn rate
- **Bank transfer (automatic):** 16.71% churn rate
- **Credit card (automatic):** 15.24% churn rate lowest churn

Insight:

Customers using electronic checks churn at nearly three times the rate of those on automatic payment methods. Automatic payments create friction to cancellation and signal a higher level of commitment to the service.

5. Churn by Internet Service

Internet service type also shows a strong relationship with churn behaviour:

- **Fiber optic:** 41.89% churn rate the highest among service types
- **DSL:** 18.96% churn rate moderate
- **No internet service:** 7.40% churn rate the lowest

Despite fiber optic being a premium service, it has the highest churn rate. This suggests that fiber optic customers may have higher expectations around speed, reliability, and value for money that are not being consistently met.

6. Churn by Senior Citizen Status

Senior citizens represent a smaller segment of the customer base but churn at a significantly higher rate:

- **Senior citizens:** 41.68% churn rate
- **Non-senior citizens:** 23.61% churn rate

Senior customers are churning at nearly double the rate of non-senior customers.

7. Churn by Tenure

The tenure analysis reveals one of the most critical findings in the entire dataset:

- Churn is dramatically highest in the first 1 to 5 months of a customer's lifecycle
- As tenure increases beyond 12 months, churn rate drops sharply and stabilises
- Long-tenured customers (beyond 50 months) have very low and consistent churn rates

Critical Insight: The first few months of a customer's journey are the highest-risk period. If the business can successfully retain customers through their first year, the likelihood of long-term loyalty increases significantly.

8. Overall Churn Distribution

The overall churn distribution confirms that the majority of customers are retained, but the proportion of churned customers is high enough to warrant serious attention:

- **Retained customers (No):** 5,174 which is 73.46% of total
- **Churned customers (Yes):** 1,869 which is 26.54% of total

A 26.54% churn rate is significantly high indicating that this business has a meaningful retention problem that requires strategic intervention.

9. Business Recommendations

Based on the analysis, the following recommendations are made:

1. Incentivise Long-Term Contracts

Since month-to-month customers churn at 42.71%, the business should offer meaningful discounts and loyalty rewards to encourage customers to commit to one or two-year contracts. Converting at least 20% of month-to-month customers to annual contracts would reduce churn significantly.

2. Promote Automatic Payment Methods

Customers on automatic payment methods (bank transfer or credit card) churn at rates as low as 15.24%, compared to 45.29% for electronic check users. The business should encourage migration to automatic payments through incentives such as a monthly discount or billing credit.

3. Invest in Early Customer Onboarding

Given that churn is highest in the first few months of tenure, the business should implement a structured onboarding programme including welcome calls, tutorials, proactive support check-ins, and early loyalty rewards to reduce churning at a early stage.

4. Investigate Fiber Optic Service Quality

Fiber optic customers have the highest churn rate despite paying a premium price. The business should conduct customer satisfaction surveys among fiber optic users to identify pain points.

5. Develop a Senior Customer Retention Programme

Senior citizens churn at 41.68% nearly double the non-senior rate. A dedicated programme offering simplified billing, dedicated support lines, and tailored pricing plans for senior customers could meaningfully reduce churn in this segment.

6. Prioritise Churn Prevention Over Acquisition

With \$2,862,926.90 lost to churned customers, the cost of inaction is significant. Retaining an existing customer is 5 times cheaper than acquiring a new one. The business should redirect a portion of its marketing budget toward retention campaigns targeting high-risk customer segments.

10. Key Findings Summary

Area	Finding	Value
Contract Type	Month-to-month has highest churn	42.71%
Payment Method	Electronic check drives churn	45.29%
Internet Service	Fiber optic churns most	41.89%
Senior Citizens	Churn at nearly double the rate	41.68%
Tenure	First months are highest risk	Month 1–5
Revenue Lost	Churned customer revenue	\$2,862,926.90
Overall Churn	Very high churn rate	26.54%

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